Charagundla Hemanth Sai

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Professional Summary

Artificial Intelligence & Data Science undergraduate with hands-on experience designing end-to-end data engineering, analytics, and machine learning solutions using PostgreSQL, Python, and Power BI for business decision support.

Education

B.E – Artificial Intelligence and Data Science Expected Graduation: 2028

Chaitanya Bharathi Institute of Technology (CBIT) CGPA: 8.80

Professional Training & Experience

Evernorth (Cigna) × CBIT AI/ML Training Program Aug 2026 – Nov 2026 (Ongoing)

- Selected for a 4-Months industry-oriented AI/ML training program conducted by CBIT with Evernorth (Cigna), focused on AI, machine learning, deep learning, generative AI, NLP, and healthcare analytics.
- Training includes hands-on exposure to PySpark, Databricks, cloud platforms, Power BI, Tableau, and healthcare-focused AI applications, with an upcoming capstone involving AI agents and small language models (SLMs).

Skills

Languages: Python | SQL | C++ | Java | R

Databases: PostgreSQL | Oracle SQL

Data Science & Machine Learning: Pandas | NumPy | Scikit-learn | XGBoost | TensorFlow | SHAP

Visualization: Power BI | Tableau | Plotly | Matplotlib | Seaborn

Tools: Git | GitHub | Streamlit | Jupyter Notebook | VS Code

Core Concepts: Statistics | Probability | Linear Algebra | Data Structures & Algorithms | OOP

Projects

SPARCS Healthcare Business Intelligence Platform GitHub PostgreSQL | SQL | Power BI

- Designed a production-style PostgreSQL healthcare warehouse integrating 16.3M+ SPARCS records across 7 years using a layered ETL architecture and Star Schema.
- Engineered a reusable multi-year ETL pipeline supporting 7 years of incremental loading with surrogate keys, Unknown Member handling, audit logging, data validation, and date dimension integration.
- Engineered an analytics layer comprising 11 SQL analytical views, 6 materialized views, 8 business reporting modules, and 8 interactive Power BI dashboards for executive, financial, clinical, operational, geographic, and trend analysis.
- Optimized analytical workloads using indexes, planner statistics, EXPLAIN ANALYZE, and materialized views while documenting the ETL architecture, KPIs, and business reporting workflow.

Medical Inventory Forecasting & Decision Support System GitHub | Live Python | Scikit-learn | SHAP

- Built an end-to-end Medical Inventory Forecasting & Decision Support System integrating data preprocessing, feature engineering, SHAP explainability, K-Means segmentation, and business decision support.
- Evaluated 14 machine learning and ensemble models, selecting a Gradient Boosting Regressor that achieved an R^2 of 0.7978, RMSE of 9.692, and MAE of 4.415 for demand forecasting.
- Implemented SHAP explainability, error analysis, and K-Means segmentation to identify demand patterns, interpret model predictions, and support procurement planning and inventory risk assessment.
- Designed and deployed an 8-page Streamlit application featuring interactive dashboards, demand prediction, recommendation modules, model comparison, SHAP explainability, inventory segmentation, and error analysis.

Additional Projects

Anime & Manga Analytics Platform GitHub | Live Python | Pandas | Matplotlib | Seaborn

- Analyzed anime and manga trends through EDA, building a 3-page Streamlit platform with Power BI dashboards and a hybrid recommendation engine using TF-IDF and cosine similarity.

Hate Speech Detection GitHub | Live Python | Scikit-learn | NLP

- Built a hate speech detection system using NLP preprocessing, TF-IDF vectorization with 5,000 features, and Logistic Regression for binary text classification.

Achievements & Certifications

- Solved 200+ problems on LeetCode, including 120+ Data Structures & Algorithms and 85+ SQL problems.
- Earned HackerRank SQL (Advanced) certification. Link

Languages

English: Proficient | **Hindi:** Proficient | **Telugu:** Native